

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP5001		
Subject Title	Semiconductor Materials and Processing		
Credit Value	3		
Level	5		
Pre-requisite / Co-requisite/ Exclusion	Nil		
Objectives	This subject aims to provide students with detailed knowledge and understanding of materials science and the processing of semiconductor materials.		
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand the fundamental electrical properties of solid crystals; (b) calculate the energy band properties of crystalline solids; (c) model the carrier properties, <i>n</i>- and <i>p</i>-type formulas, state and carrier concentrations in semiconductors; (d) study the dynamic properties of carriers such as drift, diffusion, recombination-generation, and equilibrium; (e) learn different processing techniques for semiconductor materials; and (f) understand the fabrication techniques for the integration processes.		
Subject Synopsis/ Indicative Syllabus	Fundamentals of Semiconductors: Material properties, crystal structure, and crystal growth of silicon. The Band Theory of Solids: The Kronig-Penney model, the Ziman model, the Feynman model, the effective mass, and the definition of electrons and holes. Energy Band and Carriers in Semiconductors: Bonding forces and energy bands in semiconductors, state and carrier distribution, charge carriers in semiconductors, and equilibrium carrier concentrations. Dynamic Properties of Carriers in Semiconductors: Drift, diffusion, recombination-generation, conductivity, and equation of state. Unit Process: Oxidation, photolithography, etching, doping, deposition, and planarization. Process Integration: Bipolar, CMOS, and BiCMOS technologies.		
Teaching/Learning Methodology	Lecture: Basic principles of semiconductor materials as well as their fabrication and processing especially silicon will be explained. Concrete examples will be given to illustrate the difference between the different designs of microelectronics devices. Assignment sets will be given to assess the learning progress of students. Tutorial: Students will work on problem sets in the tutorials, which provide them opportunities to apply the knowledge gained in lectures.		
Assessment Methods in Alignment with	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)

Intended Learning Outcomes			a	b	c	d	e	f
	1. Continuous Assessments (Assignments and tests)	40%	✓	✓	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓	✓	✓
	Total	100 %						
Continuous assessments, which include assignments and tests, aims at checking the progress of students throughout the course as well as assisting the students in self-monitoring their performance. The examination will use to assess the knowledge acquired by the students, as well as to determine the extent to which they have achieved the intended learning outcomes.								
Student Study Effort Expected	Class contact:							
	▪ Lecture					33 Hrs.		
	▪ Tutorial					6 Hrs		
	Other student study efforts:							
	▪ Reading and self-study					81 Hrs.		
	The total student study effort					120 Hrs.		
Reading List and References	L. Solymar and D. Walsh, Lectures on the electrical properties of materials, 3 rd Ed., Oxford University Press, 1984.							
	S.M. Sze, Kwok K. Ng, Physics of Semiconductor Devices, John Wiley & Sons, 2006 (https://onlinelibrary.wiley.com/doi/book/10.1002/0470068329).							
	Gary S. May and Costas J. Spanos, Fundamentals of Semiconductor Manufacturing and Process Control, John Wiley & Sons, 2006							
	A. Rockett, The Materials Science of Semiconductors, Springer, 2008.							